Oracle
Retrained on Homology-Restricted MegaScale
Zero-shot
Supervised, not trained on Megascale
Rectified gain @ k=24 (kcal/mol)
Predicted most stabilizing per domain
0
1
10
3.28 1.80 6.68 6.09
1.21 1.01 4.66 3.29
1.57 0.96 4.64 3.70
0.76 0.78 2.75 2.60
1.04 0.56 4.21 2.91
1.35 0.69 5.02 2.55
1.05 1.25 1.31 1.32
1.44 1.77 1.65 1.18
0.40 0.24 0.82 0.39
1.07 0.30 3.14 1.19
0.99 0.50 2.86 1.89
4.00 2.67 8.61 8.73
2.86 2.65 9.98 8.52
3.11 2.34 9.66 8.89
2.75 3.42 8.38 9.44
3.15 2.10 10.80 7.95
3.14 1.99 12.17 6.70
5.09 3.85 8.15 6.91
6.33 4.48 8.45 4.73
6.64 5.58 6.21 5.58
5.55 1.42 11.18 5.85
3.42 3.15 8.85 8.58
Top 24 per domain
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
MutateEverything st1
ThermoMPNN
SPURS
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta
Δ hydrophobicity ≤ 0 & relative ASA ≤ 0.5
Δ hydrophobicity > 0 & relative ASA ≤ 0.5
Δ hydrophobicity ≤ 0 & relative ASA > 0.5
Δ hydrophobicity > 0 & relative ASA > 0.5

Top 24 per Homology-Restricted MegaScale test domain (33 domains with the feature, 35,295 single mutants); gain = sum of max(ddG, 0) over the picks predicted stabilizing (> 0); boxes: interquartile range with median line; dots: domains; annotations: mean over domains; dashed lines: oracle mean.